Full-Stack Coding Assignment: Kanban Task Board

Overview

We'd like you to build a simple Kanban board application for managing tasks. The application should allow users to create, view, update, and delete tasks, and move them between columns (e.g., To Do → In Progress → Done).

Your solution should demonstrate:
- Backend API design and implementation in .NET Core
- Frontend implementation in React or Vue.js
- Clean architecture, coding practices, and maintainability
- Proper use of state management and component design
- UI/UX usability

Requirements

Backend (.NET Core 6+)

- - Implement a REST API with endpoints for: Create, Get all, Update, Delete tasks
- - Each task should have: Id, Title, Description, Status (To Do, In Progress, Done), CreatedAt, UpdatedAt
- - Store data in an in-memory database (e.g., EF Core InMemory provider). (Bonus: use SQLite or MongoDB)
- - Follow REST best practices (HTTP verbs, proper status codes, etc.).

Frontend (React or Vue.js)

- - Implement a Kanban-style board UI with three columns: To Do, In Progress, Done
- - Features: Add new task, Edit task, Delete task, Drag and drop between columns
- - The frontend should consume the API you build.
- - Use state management (React hooks/Context/Redux or Vuex/Pinia).
- - UI should be clean and usable (not necessarily pixel-perfect).

Expectations

- - Write clean, readable, and maintainable code.
- - Use Git for version control. Share a public repo (GitHub/GitLab/Bitbucket).
- - Include setup instructions in a README.md (how to run backend & frontend).
- - Timebox: spend no more than 4 days on this exercise.

Evaluation Criteria

We will review your submission based on:

Backend

- - API design, architecture, async/await usage, validation, error handling

Frontend

- - Code structure, state management, CRUD handling, drag/drop implementation, UI clarity

General

- - Code readability, error handling, Git commit history, documentation clarity